

Getting started with ST25RN300 evaluation board

Introduction

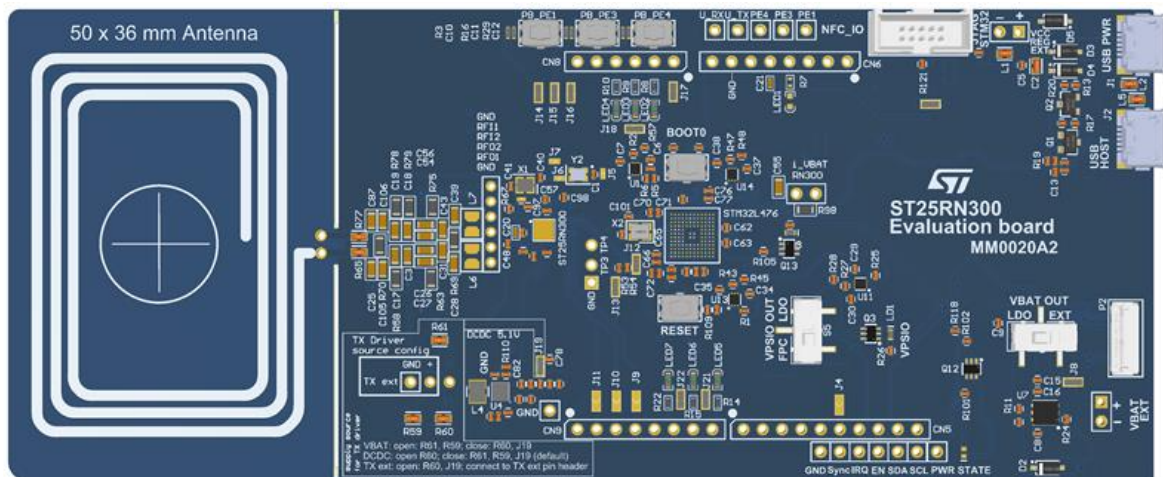
The ST25RN300 evaluation board supports a wide range of contactless applications. It comes as a complete development kit, ready for immediate testing with EMVCo reduced range 2 (EMVCo RR2 v3.2a) and other contactless standards.

The reference design includes the ST25RN300 near field communication (NFC) reader device connected to a 50 x 36 mm antenna integrated on the board. The three-turn RF antenna detaches easily along a tearing edge to meet diverse application requirements.

The onboard STM32L476 microcontroller manages communication between the PC host and the ST25RN300. It bridges USB to I²C communication and relays the near field communication controller interface (NCI) protocol. The platform also supports interconnectivity with other host devices, such as embedded microcontroller units (MCU) or Android application processors, via a dedicated I²C pin header or a specific 25-pin flexible printed circuit (FPC) cable. This expands its versatility across various development environments.

The evaluation board offers multiple power supply options for both the device core and the radio frequency (RF) driver. This ensures flexibility and reliable performance for a wide range of use cases.

Figure 1. ST25RN300 evaluation board PCB



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1 General information

This document applies to Arm®-based devices. This document provides information on initial setup and guides the user through the various blocks of the demoboard.

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1.1 Reference documents

Table 1. Reference documents

N°	Reference
[1]	Layout recommendations for the design of boards with ST25RN300 (AN6367)
[2]	Guidelines for ST25RN300 NCI reader antenna and RF configuration (AN6356)

2 Key features

- Based on the ST25RN300 NFC controller interface (NCI) reader for payment, access control, and consumer applications.
- Reader/Writer for NFC Forum type 1, 2, 3, 4, and 5 tags.
- Card emulation mode in ISO/IEC14443A/B and FeliCa™.
- MIFARE® Classic.
- EMVCo L1 stack integrated in firmware.
- RF transmitter supply up to 2.2 W.
- Multiple TX driver supply options.
- Dynamic Power Control (DPC) and Active Wave Shaping (AWS).
- 50 mm x 36 mm, removable three-turns antenna.
- Comprehensive PC user interface for device configuration operation.
- Host interface via a USB, I²C pin header, or FPC cable.

3 Quick start

To evaluate and test the ST25RN300 evaluation board, begin by installing the NFC explorer 2-ST25RN GUI software on your PC. Next, connect the evaluation board to the PC using a USB cable via the USB port labeled **USB host**. Once the **Connect** button in the software is clicked, the board is detected automatically, enabling immediate testing and configuration of its various functionalities.

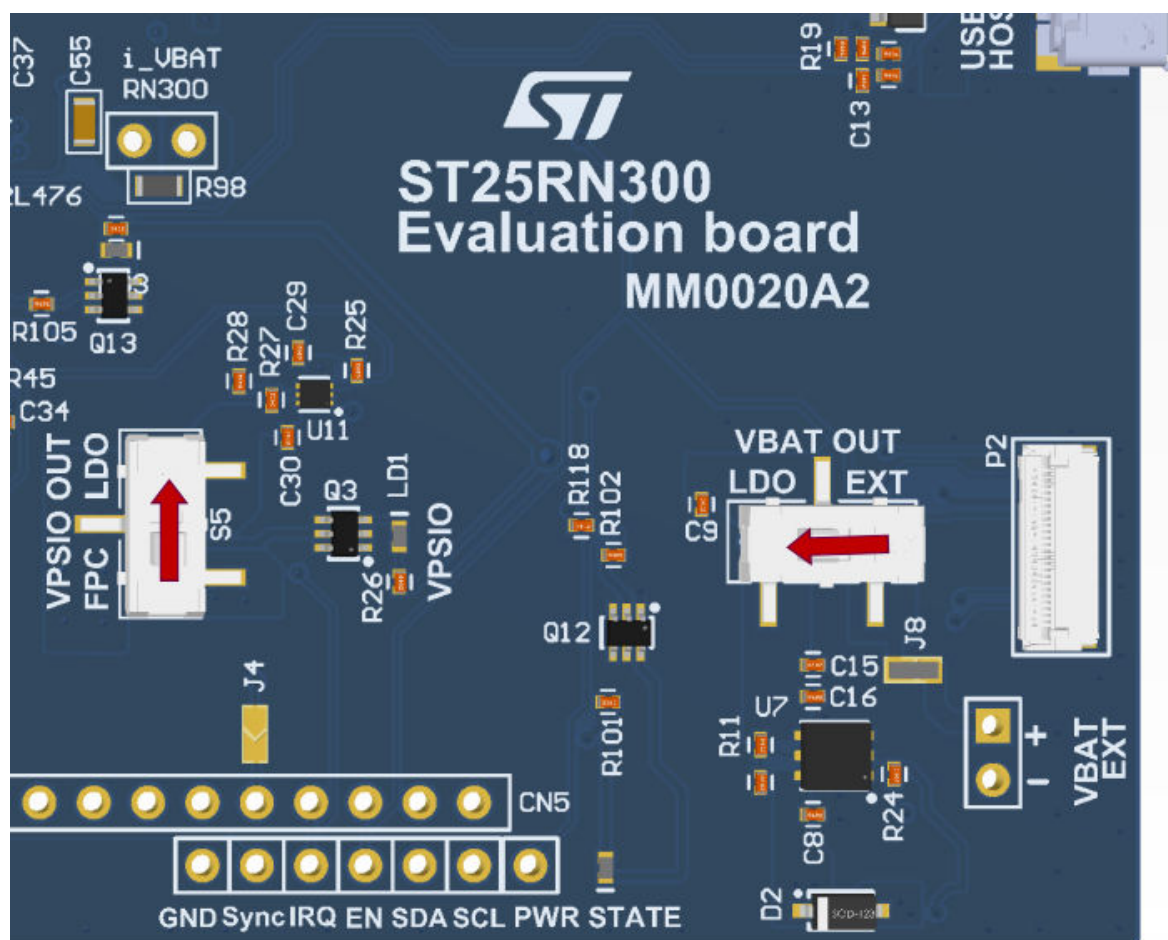
3.1 System requirements

- Windows® PC (Windows 10 or higher).
- USB port (USB host) capable of delivering at least 500 mA at a 5 V supply.
- For additional power supply to the board, a second USB power cable (labeled USB PWR) can be connected.
- Micro USB cable.

3.2 Software installation and first run

Before starting to use the evaluation board, verify that the two slide switches are set to the correct positions as indicated in [Figure 2](#). Both switches must be set to the LDO position. The switches determine the power source for the ST25RN300, which can be either USB or an external supply. More information is available in [Section 4.4](#).

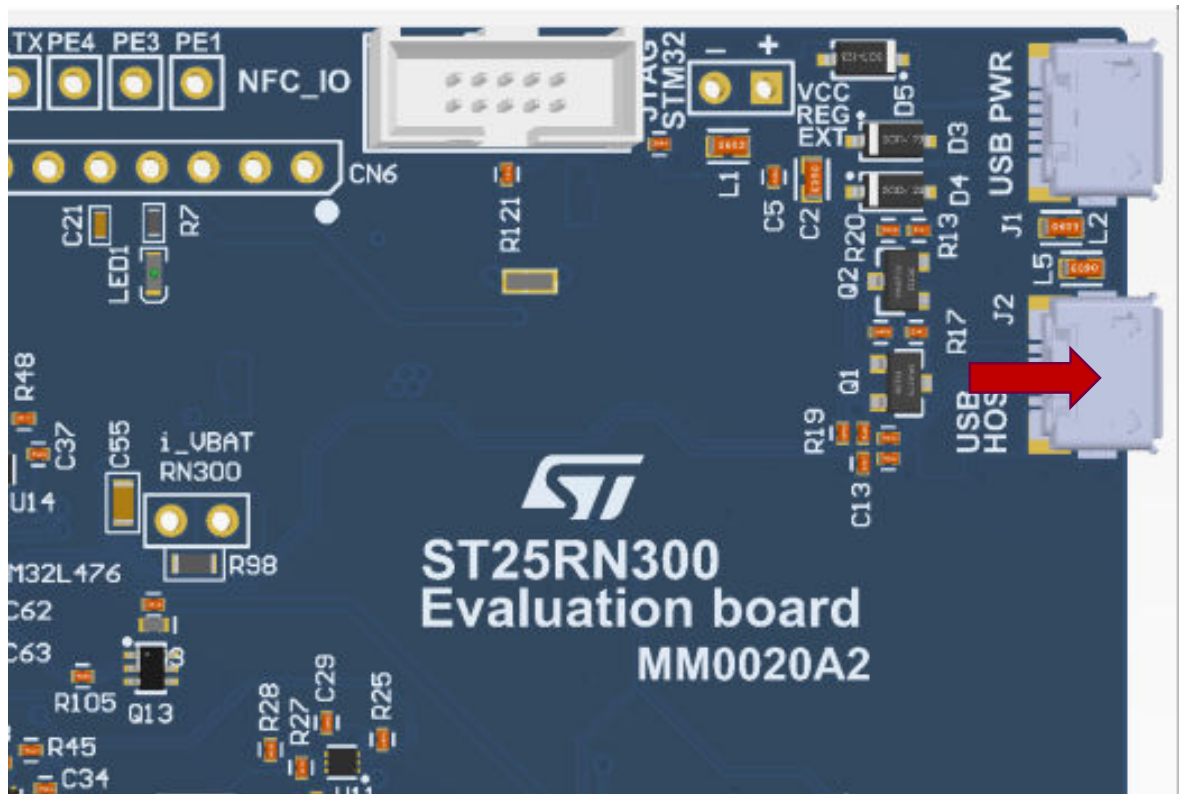
Figure 2. Switch configuration



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- Connect the micro-USB cable to the USB port labeled **USB host** to power and enumerate the board. It is important to connect the USB to the correct connector, as shown by the arrow in Figure 3.

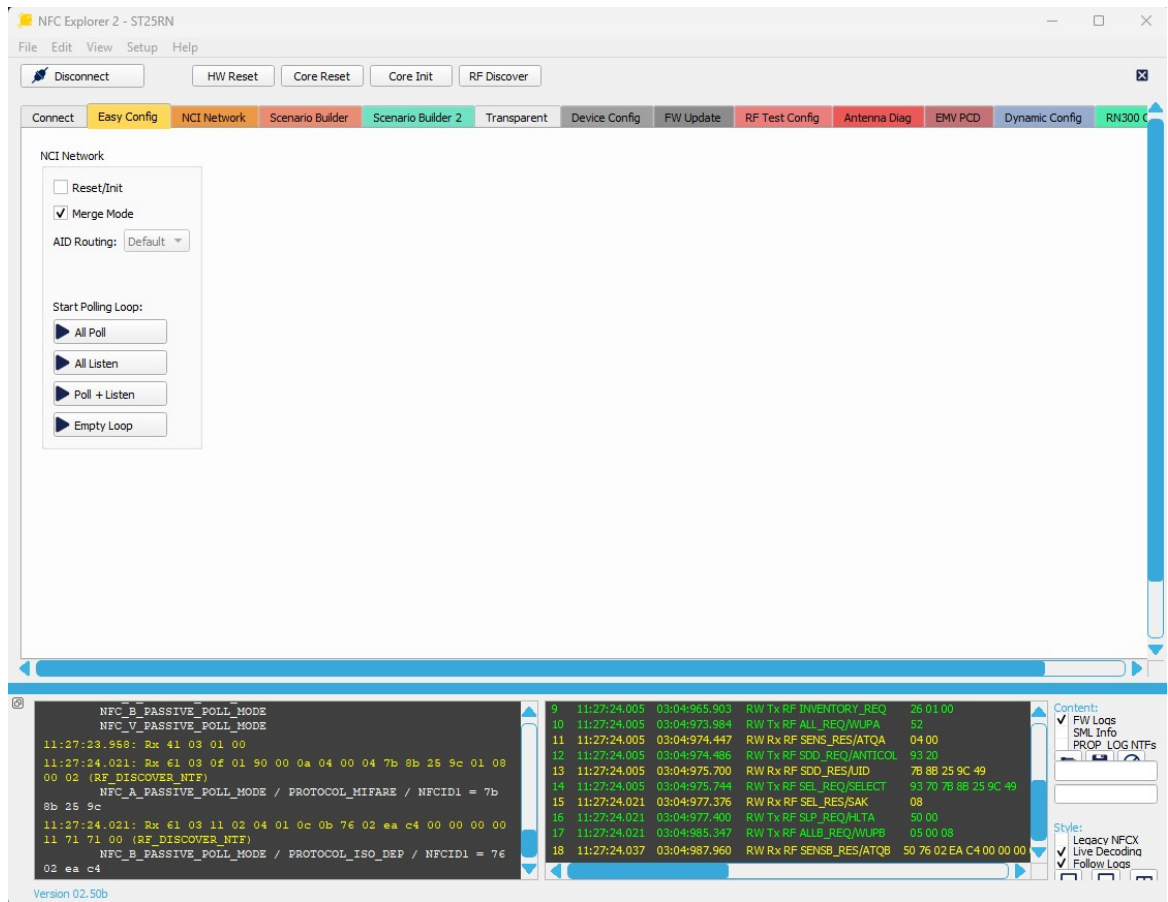
Figure 3. USB host connection



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- Start the NFC explorer 2-ST25RN GUI.
- In the top section of the GUI, click the **Connect** button.
- Go to the **Easy Config** tab and click on the **All Poll** button to initiate polling for NFC-A/B/F/V tags.
- Present a contactless NFC-A/B/F/V card on the reader and observe the exchange in the log window, which confirms the card is detected and communication is established successfully.

Figure 4. NFC explorer 2 GUI windows

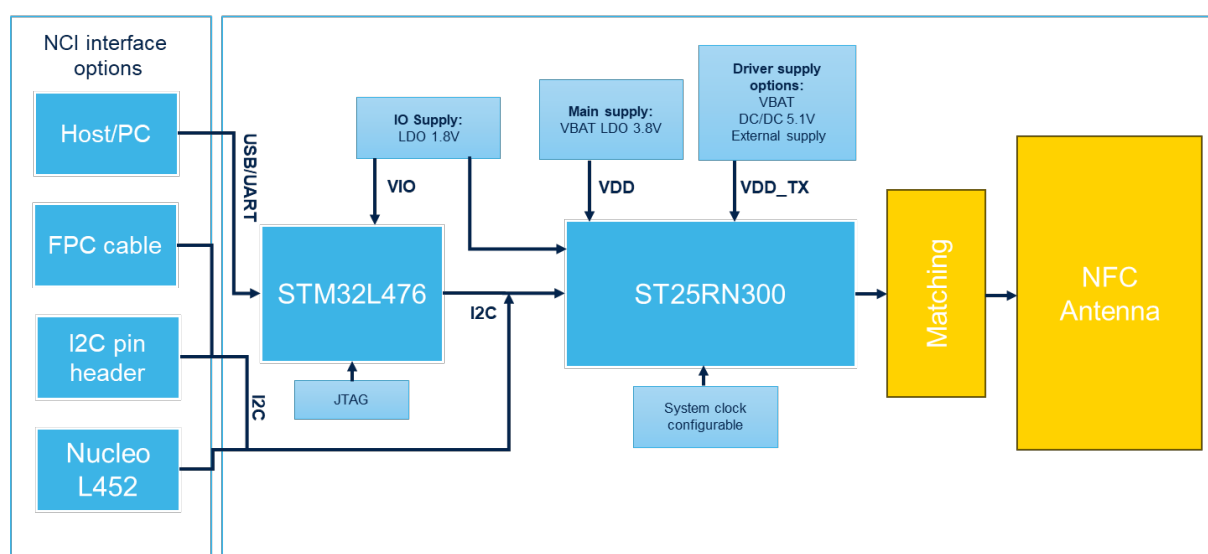


4 General hardware description

Figure 5 presents the high-level block diagram of the ST25RN300 evaluation board. The ST25RN300 controller complies with the NCI specification, and its embedded firmware is securely preprogrammed during production. The NCI interface is configured by default via the onboard switches to communicate with the PC as the host device. In this setup, the STM32L476 microcontroller manages the NCI communication over the physical I²C layer. From the PC host perspective, the STM32 appears as a USB/UART interface, which can be accessed using the NFC explorer software. The evaluation board offers a flexible power supply architecture, which is detailed in the following subsections.

Figure 5. ST25RN300 evaluation board block diagram

ST25RN300 - Evaluation board block diagram



4.1 Board layout

The ST25RN300 evaluation board is a 6-layer stack with a high-Q antenna measuring 50 x 36 mm. The antenna is attached to the mainboard but can be detached easily if a custom antenna needs to be connected. The top layer contains all active components related to the ST25RN300 and the STM32L476 microcontroller, as detailed in the following subsection. The bottom layer of the PCB is a component-free ground plane, providing a secure mechanical base when placed on a desk. Additional details on the board stacking are available in [1].

Figure 6. Top layer layout

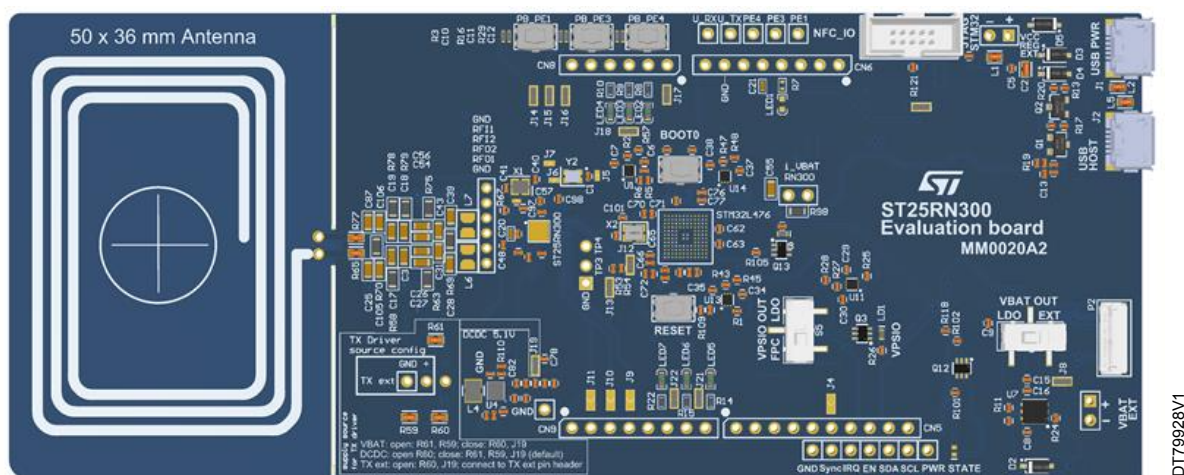


Figure 7. Bottom layer layout



4.2 ST25RN300

The minimum schematic and component details for the ST25RN300 are provided in [1]. This section focuses on the implementation specifics of the evaluation board.

Although the ST25RN300 comes with a preprogrammed firmware embedded during production, it requires an appropriate register configuration tailored to the specific use case of the board. The configuration file includes settings for the oscillator source, driver supply, receiver, waveform shaping, card mode operation, and others.

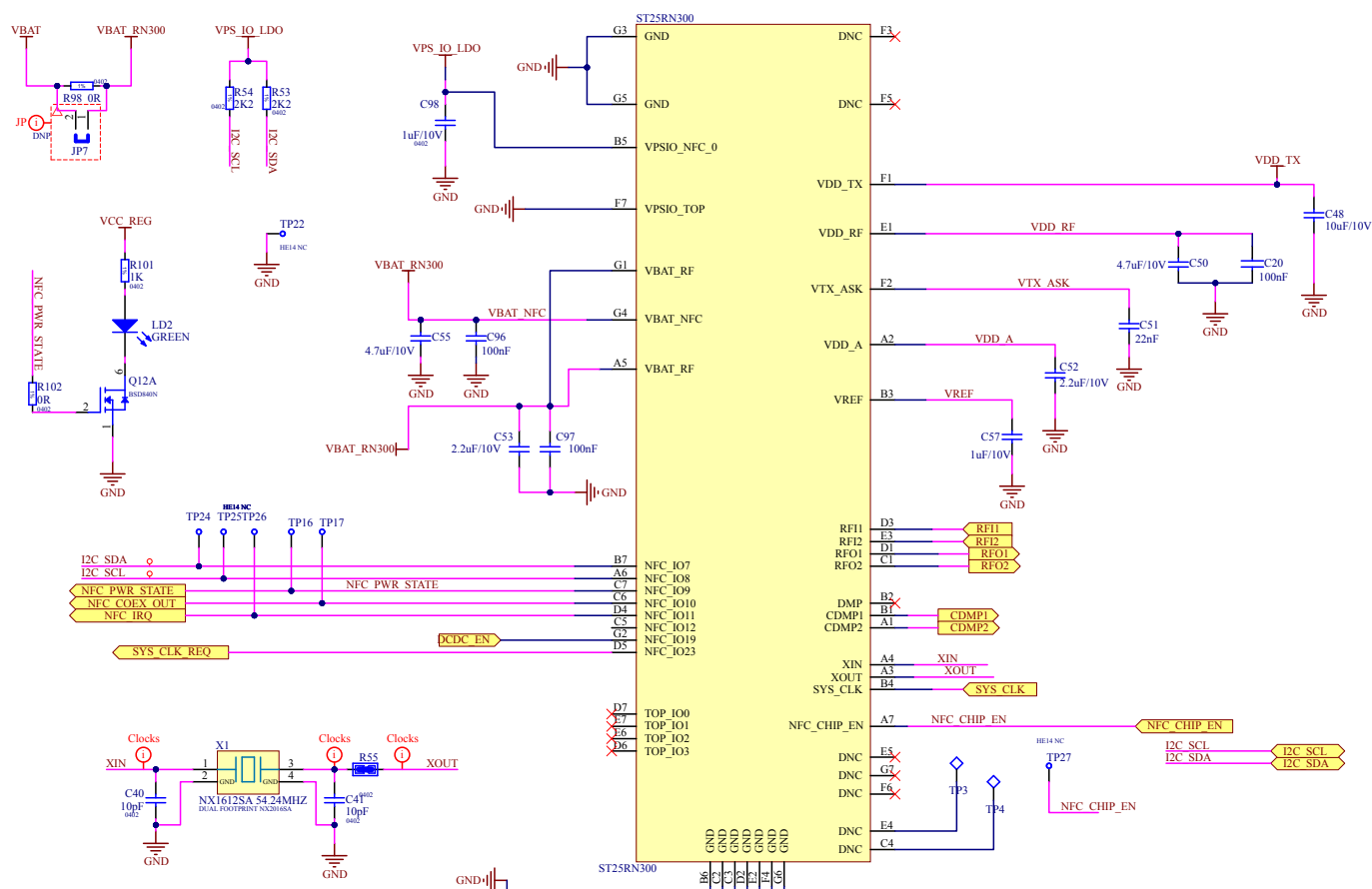
The evaluation board is shipped with a default configuration optimized primarily for EMVCo RR2 operation at a basic data rate of 106 kbit/s. However, the demo board can also be used without modification in card emulation mode. The operating mode can be selected and adjusted using the NFC explorer tool.

The status of the configuration file on the ST25RN300 can be viewed using the **Connect** tab in the NFC explorer tool. The CRC value displayed in the tool indicates whether a configuration file has been successfully loaded. Configuration files can be loaded and saved directly through the NFC explorer interface.

The ST25RN300 schematic on the demo board is similar to the one referenced in [1], with a few measurement points added and the ability to switch between an external reference clock and a crystal oscillator. The default configuration uses the 54.24 MHz crystal oscillator.

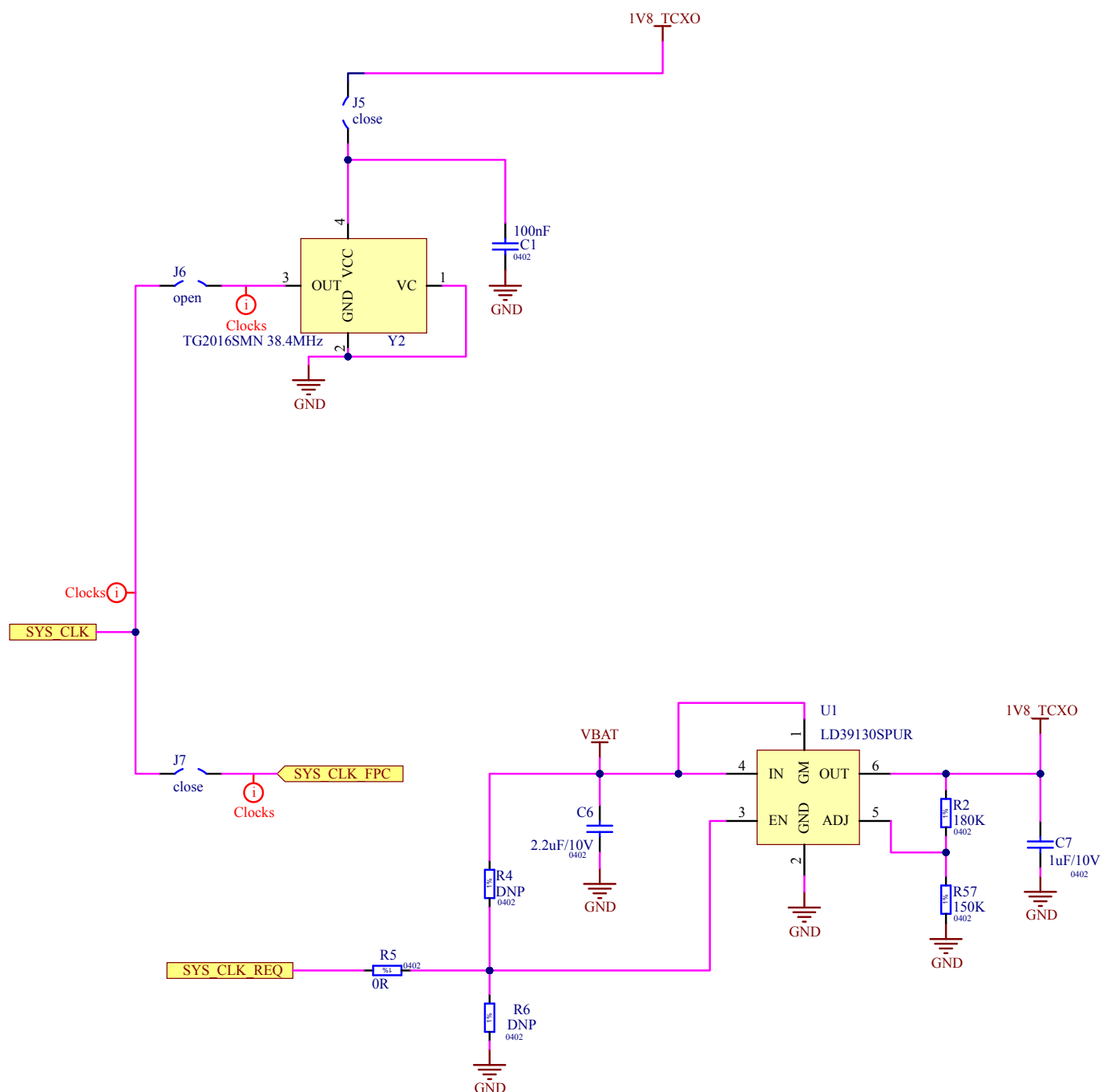
Note: An incorrect clock source configuration in the NFC explorer leads to a connection error.

Figure 8. ST25RN300 evaluation board schematic



The external reference clock is provided by a 38.4 MHz source and must be selected accordingly in the NFC explorer tool. Ensure that jumpers J5 and J6 are closed to enable the power supply to the reference clock and to route the signal to the ST25RN300. Other reference clock frequencies and parameters can be used as specified in the device datasheet.

Figure 9. External clock configuration

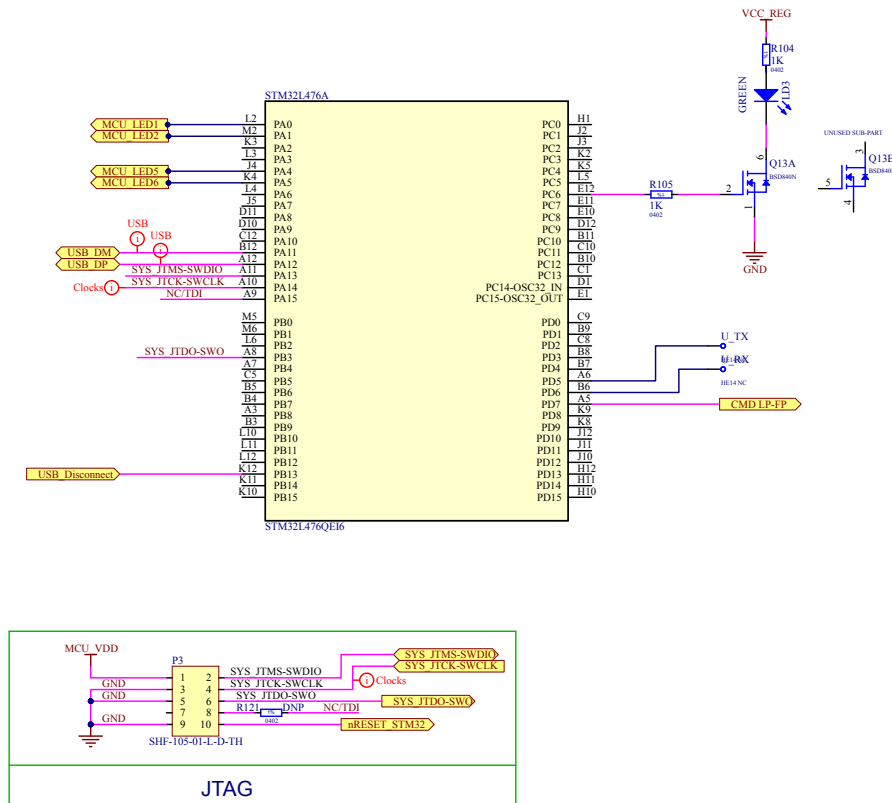


4.3 STM32L476 microcontroller

The STM32L476 MCU acts as a relay for the NCI communication between the device host and the ST25RN300 via I²C. This link is established with the SML firmware embedded in the MCU. This firmware can be updated directly via the NFC explorer tool or through the dedicated JTAG connector. An external ST-LINK is required to establish the connection between the JTAG and the MCU.

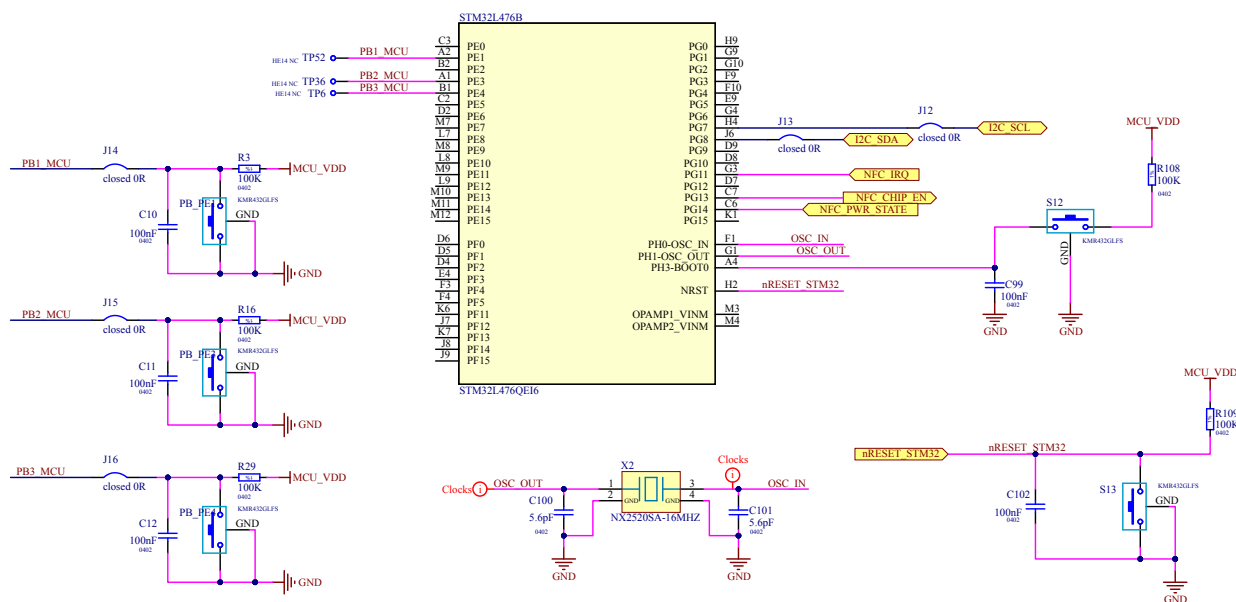
The MCU UART on pins PD5 and PD6 is routed to a dedicated pin header on the evaluation board.

Figure 10. STM32L476 schematic

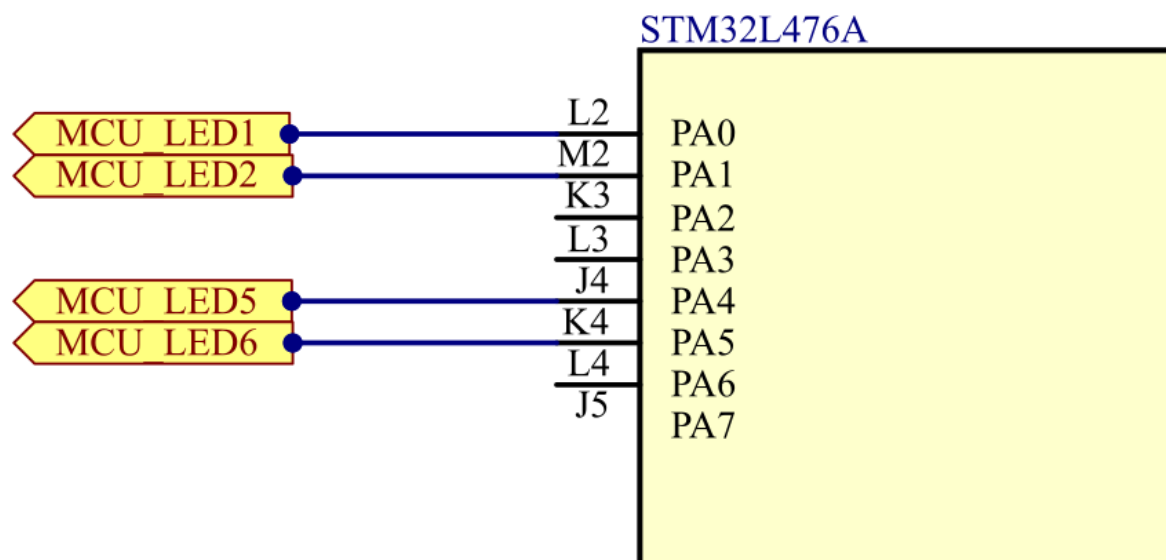


The MCU block also provides access to five pushbuttons, two of which are dedicated to entering bootloader mode. These buttons are labeled **Reset** and **BOOT0** on the board. To enter bootloader mode, press and hold the **BOOT0** button while briefly pressing and releasing the **Reset** button.

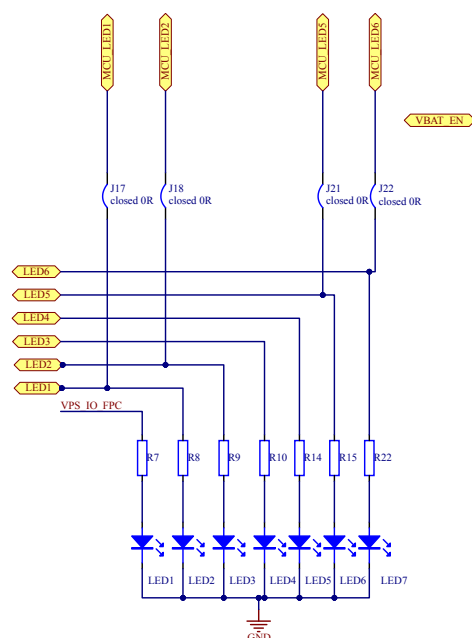
The remaining buttons, labeled PB1 to PB3_MCU, can be freely configured to trigger specific actions within custom host firmware.

Figure 11. MCU pushbutton connection diagram


Additionally, four LEDs are connected to MCU pins PA0, PA1, PA4, and PA5 and can be configured as needed.

Figure 12. LED connections to MCU


MCU_LED1, MCU_LED2, MCU_LED5, and MCU_LED6 are routed to LED2, LED3, LED6, and LED7, according to the wiring diagram in [Figure 13](#).

Figure 13. MCU LED wiring diagram


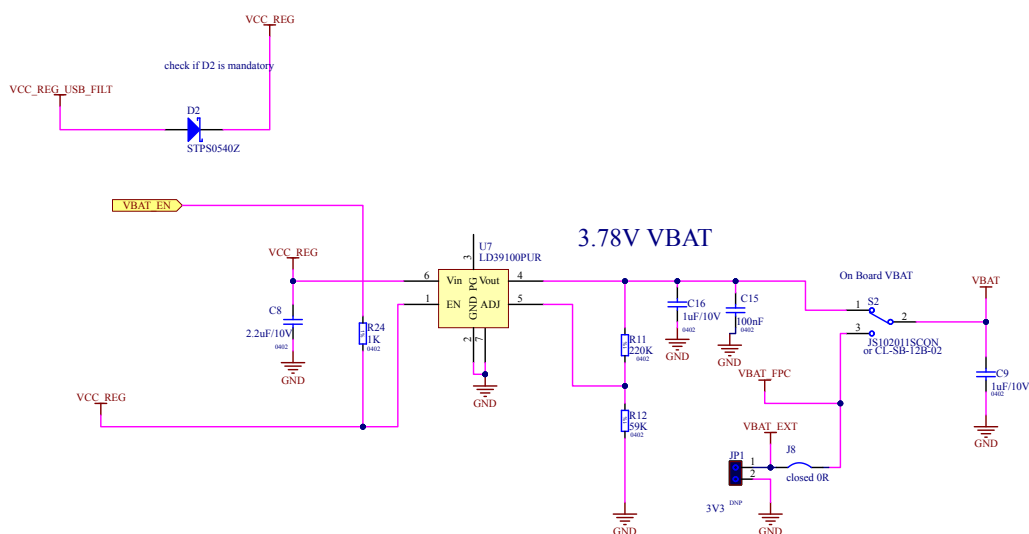
4.4 Power supply options for the PCB

By default, the ST25RN300 evaluation board is powered via the USB power line, with the two slide switches set to LDO. In this mode, all LDO regulators and the onboard DC-DC converter are supplied by the 5 V USB line. Connect the micro USB port labeled USB HOST to the PC to enable host communication and simultaneously provide power to the board. The micro USB port labeled USB PWR serves solely as a secondary power source for applications with higher power demands.

4.4.1 VBAT supply

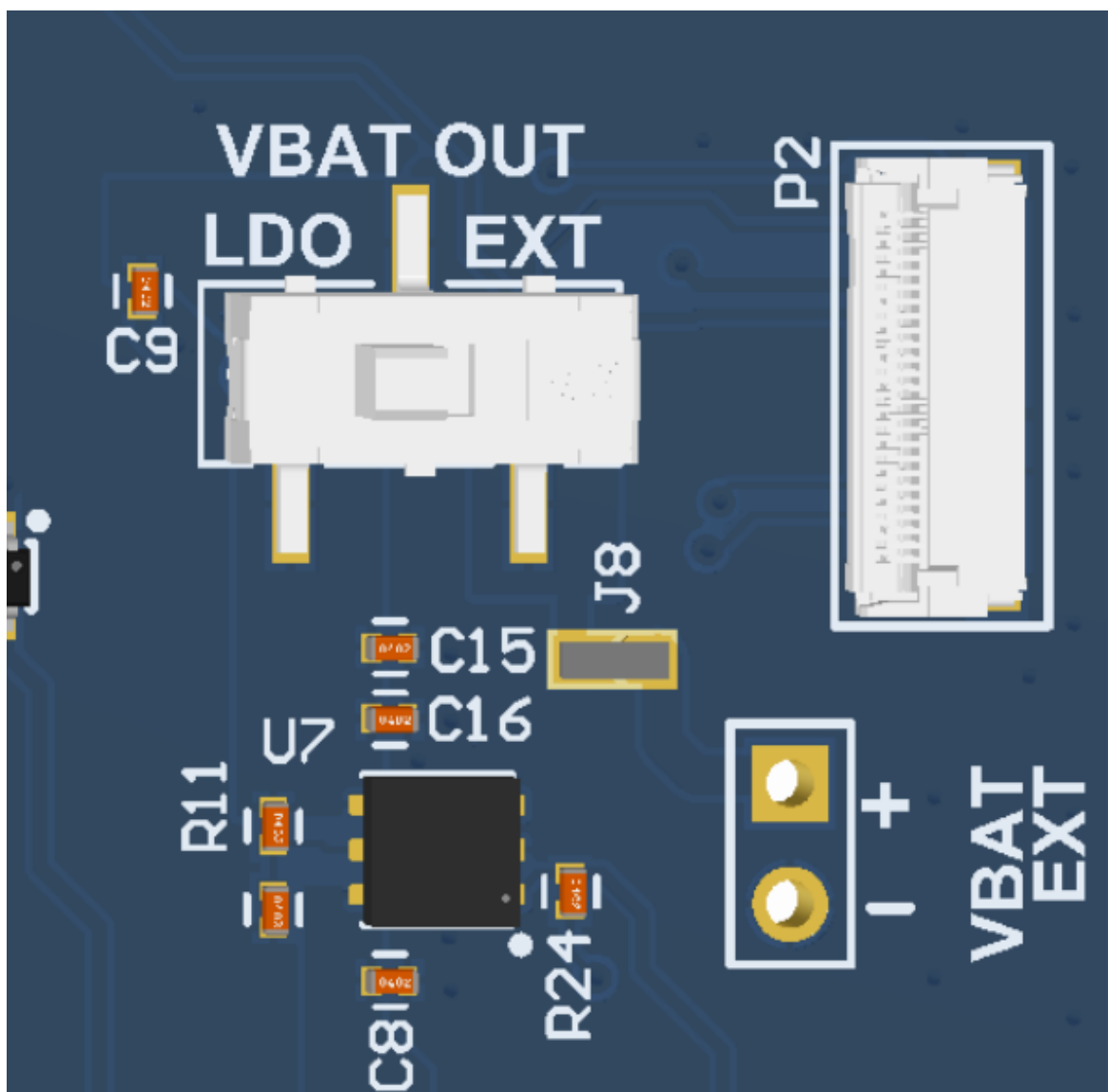
The LD39100PUR is an adjustable LDO regulator that provides a VBAT voltage of 3.78 V to the main supply of the ST25RN300. This voltage level simulates the nominal voltage of a typical battery-powered device. The VBAT voltage can be adjusted via the resistive divider formed by R11 and R12 on the LD39100 output (VOUT) or supplied externally through the VBAT_EXT input.

Figure 14. VBAT source selection schematic



The ST25RN300 pins G1, G4, and A5 are the main power supply inputs for the device, referred to as VBAT_RN300 in the schematics. The source of VBAT_RN300 must be selected using the 2-way slider switch labeled VBAT_OUT. In the "LDO" position, the supply source is routed from the USB connection. In the "EXT" position, the supply source is routed from the FPC connector or the external pin header.

Figure 15. VBAT_OUT power source selection switch

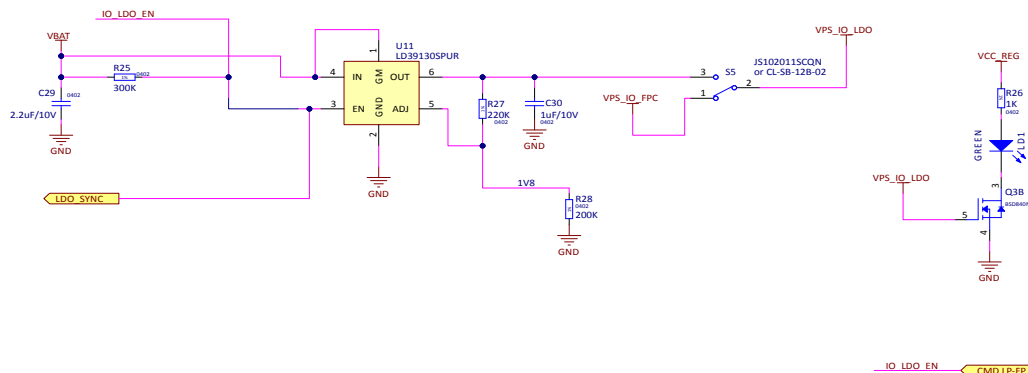


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Further information on power supply and component selection is available in [1].

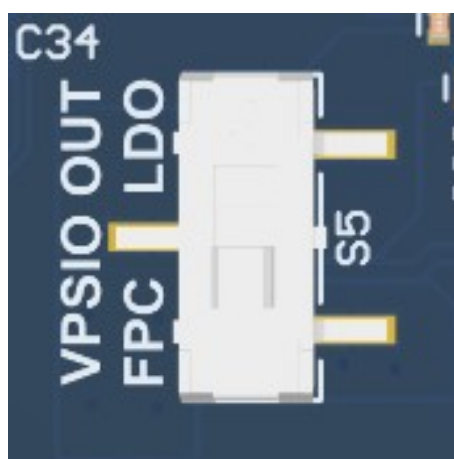
4.4.2 I/O supply

The I/O supply for both the ST25RN300 and the onboard STM32L4 MCU is derived by default from the LD39130S at 1.8 V. The input to this regulator is the VBAT voltage source.



The slide switch must be set to LDO if the I/O voltage is derived from VBAT and USB usage. The I/O supply can also be sourced from the FPC cable by switching the slider to FPC.

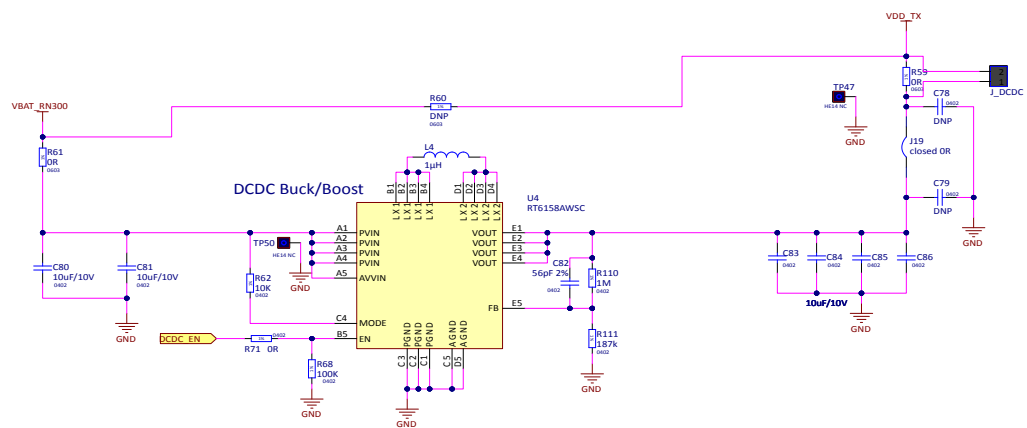
Figure 16. VP5IO_OUT source selection switch



4.4.3 ST25RN300 driver supply and onboard DC-DC

The ST25RN300 driver supply is provided by an internal LDO, powered either from VDD_TX or VBAT. The VDD_TX input can be supplied with up to 5.5 V from an external DC-DC converter. The VDD_RF output, generated by the ST25RN300 internal LDO, powers the driver stage. The onboard adjustable DC-DC buck/boost converter, RT6148AWSC, can deliver up to 5.1 V, which is the default configuration on the demo board. The DC-DC converter input is connected to the VBAT_RN300 line. This DC-DC converter can be disabled via software by controlling the enable pin (EN) of the RT6148AWSC, allowing the RF drivers to operate solely from the available VBAT voltage.

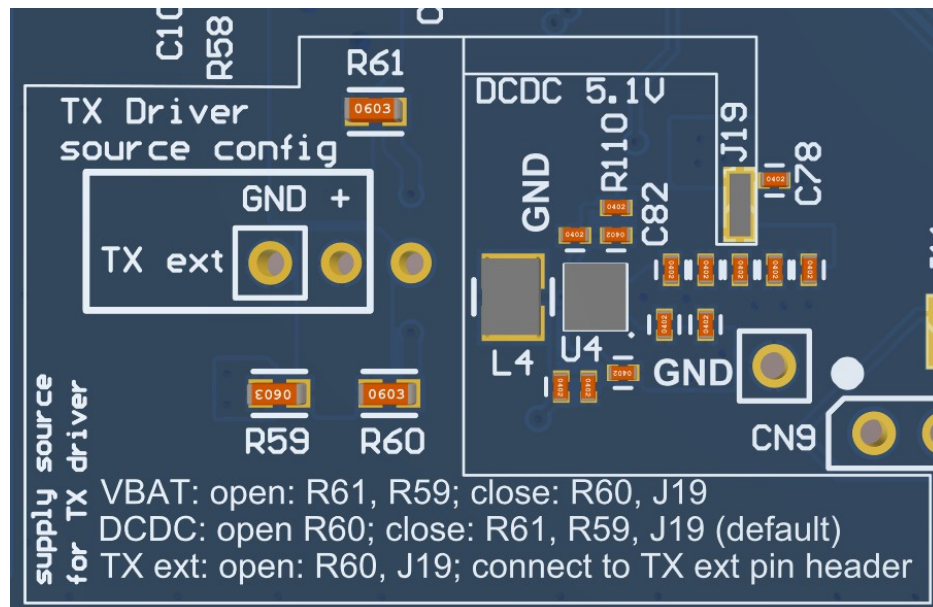
Figure 17. VDD_TX power supply scheme



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The following layout snippet shows the possible jumper configurations for the source of VDD_TX, which are named VBAT, DC-DC, or external supply.

Figure 18. TX driver supply source jumper configuration



The default configuration uses the onboard DC-DC converter, which outputs 5.1 V set by the voltage divider resistors R110 and R111. To enable the DC-DC converter, resistor R60 must be open, while R61, R59, and jumper J19 must be closed.

The VBAT supply can be selected via software configuration. The DC-DC converter can also be bypassed in hardware by opening R61 and R59 and closing R50 and J19.

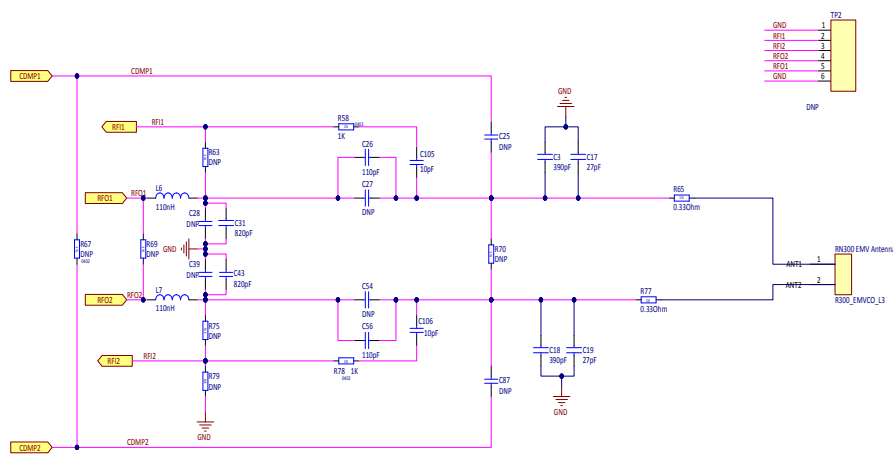
If a driver supply voltage higher than 5.1 V is required, it must be supplied externally through the TX_EXT pin header. In this case, pads R60 and J19 must be opened.

In summary, this flexible power architecture allows the board to be powered from USB or an external supply. The ST25RN300 driver supply can be sourced from the DC-DC converter, the onboard VBAT supply, or an external power input.

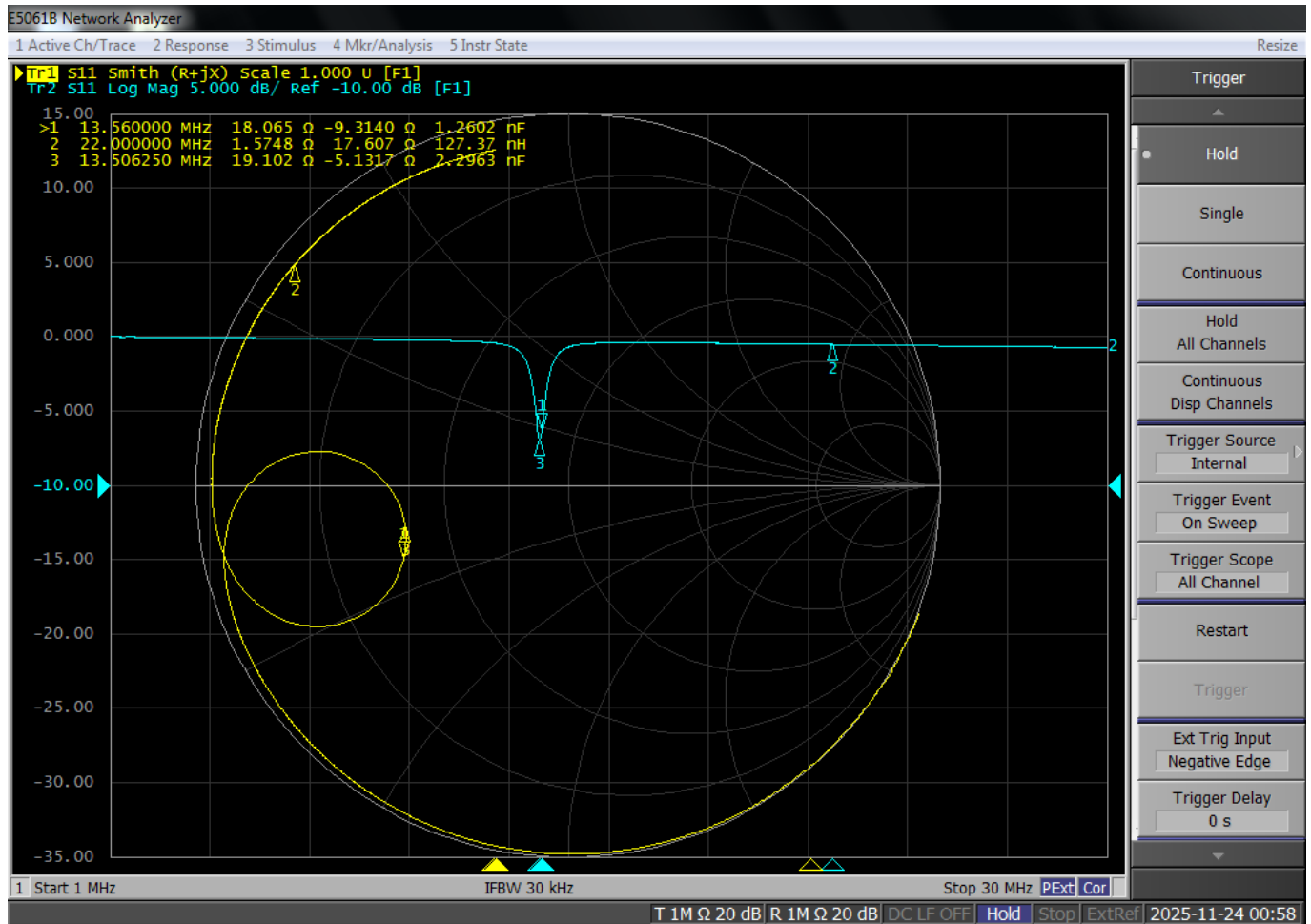
4.5 Antenna matching

The antenna matching is optimized to comply with EMVCo reduced range 2 analog operation at a basic data rate of 106 kbit/s for NFC-A and NFC-B. The matching impedance of the ST25RN300 evaluation board is approximately $18-j9\ \Omega$, ensuring efficient power transfer and fast modulation transients.

Figure 19. Antenna matching schematic



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Figure 20. ST25RN300 evaluation board matching impedance


Additional information on antenna matching is available in [2].

4.6 Peripheral connection

The board offers multiple pin headers to enable connections in two ways.

4.6.1 Connection to the device host via a dedicated I²C pin header

One method to connect externally a device host to the ST25RN300 is through the dedicated I²C pin header located on the side of the demo board. Wires should be soldered to the individual signal pins, and the two switches must be set to FPC and EXT, respectively.

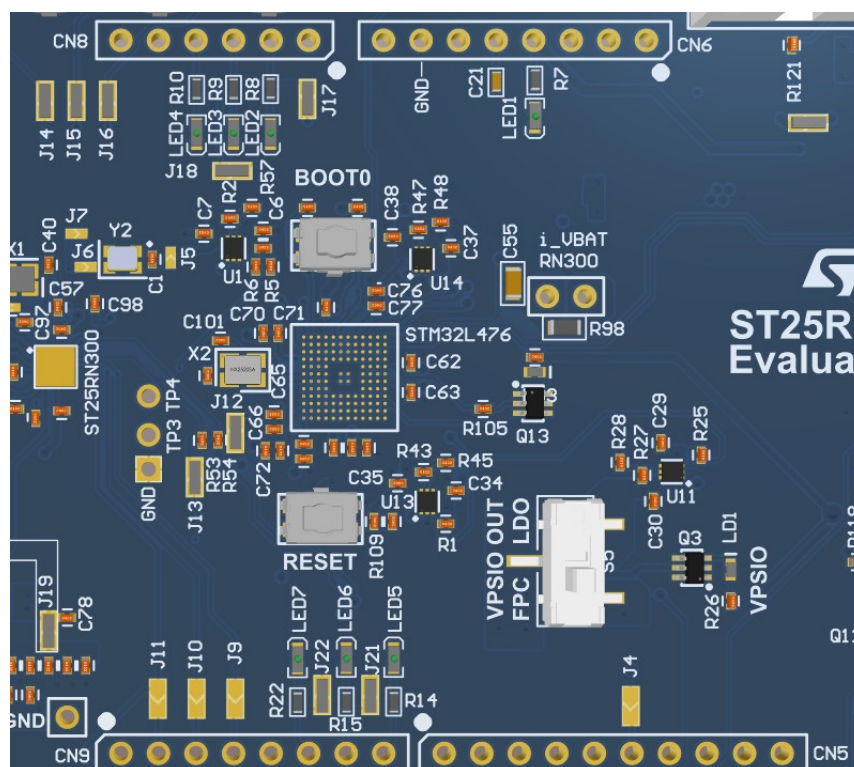
Figure 21. External host connection via I²C pin header


4.6.2 Connection to NUCLEO-L452RE-P

The pin headers CN5, CN6, CN8, and CN9 are positioned to align precisely with the Nucleo shield, enabling seamless integration of the ST25RN300 evaluation board with the Nucleo board as a host controller. However, due to the 1.8 V I/O voltage limitation of the ST25RN300 device, only the I/O voltage of the NUCLEO-L452RE-P can currently be easily reconfigured to match the evaluation board I/O voltage levels.

Further implementation details are provided once this solution is fully validated.

Figure 22. Nucleo shield pin headers for ST25RN300 integration



Revision history

Table 2. Document revision history

Date	Revision	Changes
02-Dec-2025	1	Initial release.

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